



The bridge to possible

SRv6 Blueprint

(Addressing)

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- uSID Locator Format
- GIB/LIB
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- Large Network
- Loopback and Interfaces

SRv6 Addressing Requirements

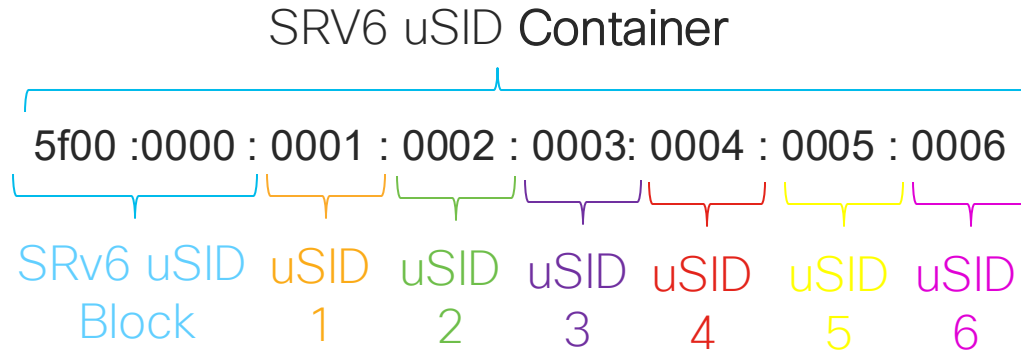
- Simple
 - Operationally
- Hierarchical
 - Summarization
- Extensible
 - Allow growth
- TE Efficient
- ULA/GUA/5F
- SRv6 addressing is independent of loopback/link addressing!!

ULA-GUA-5F

- ULA Unique Local FC00:/7
 - Most deployments
 - Very flexible
 - Inherent Security (Still Infrastructure ACL required)
- GUA Global Unicast
 - Assigned by authority (IANA , AFRNIC, APNIC, ARIN, RIPE)
 - Consumes significant portion of public address space
- 5F00::/16- RFC9602
 - Same properties as ULA

uSID Format

- Currently all deployments and interop uses “F3216”
 - 32 bits uSID block
 - 16 bits uSID



TE Compression Efficiency for uSID

- uSID is the most efficient TE compression schema!
 - With proper addressing plan ☺

Overhead

SA:5f00:0:7543::1
DA:5f00:b5:6::
NH:RH

Type: 4 (SRH)
NH:IPv4 SL:5
Segment List:
[0]: 5f00:b0:1::
[1]: 5f00:b1:2::
[2]: 5f00:b2:3::
[3]: 5f00:b3:4::
[4]: 5f00:b4:5::

SIDs from different blocks!

5 Containers

Very Efficient

SA:5f00:0:7543::1
DA:5f00:b0:1:2:3:4:5:6
NH:IPv4

All SIDs from single block!

1 Container

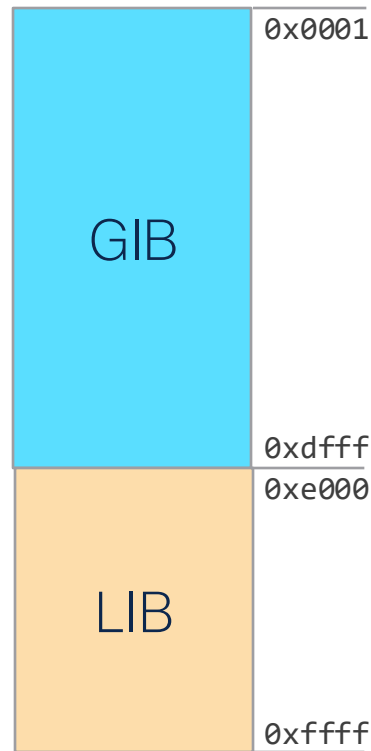
Addressing plan must minimize number of blocks any TE policy will cross!!!

“Small Network” = Single Block

- How much devices fits single block?
- Each SRv6 device must have single uSID assigned
- uSIDs is 16bits = 65535 unique values
- But....

Sets, Global ID Block (GIB), Local ID Block (LIB)

- GIB= Global ID Block ~**57k** values
 - Locator=5F00:b0:**1**::/48
 - uN=5f00:b0:**1**::
 - Represents shortest path to a node
- LIB=Local ID Block ~8k values
 - uA=5F00:b0:**1**:**e000**::
 - uDT4=5F00:b0:**2**:**e001**::
 - All local functions



Summarization Inefficiencies

- Summarization can be performed on area (level boundaries)
- Each area have certain number of devices
- Unlikely to match binary boundaries
- Area with 350 devices?

5f00:b1:0100::/39 - 512 devices - 162 address unused

- Single Block can be used for networks ~<35k SRv6 devices

Area addressing

- Network has areas (levels)
 - Area size varies
 - (100s-1000s devices)
- How to allocate address ranges to the areas?
 - Do not waste addresses
 - Simplicity

Area addressing - Example

3 areas A1:350, A2:2100, A3:750 devices

Simple

A1: /36₍₄₀₉₆₎

A2: /36₍₄₀₉₆₎

A3: /36₍₄₀₉₆₎

Not Efficient

Efficient

A1: /40₍₂₅₆₎ /41₍₁₂₈₎

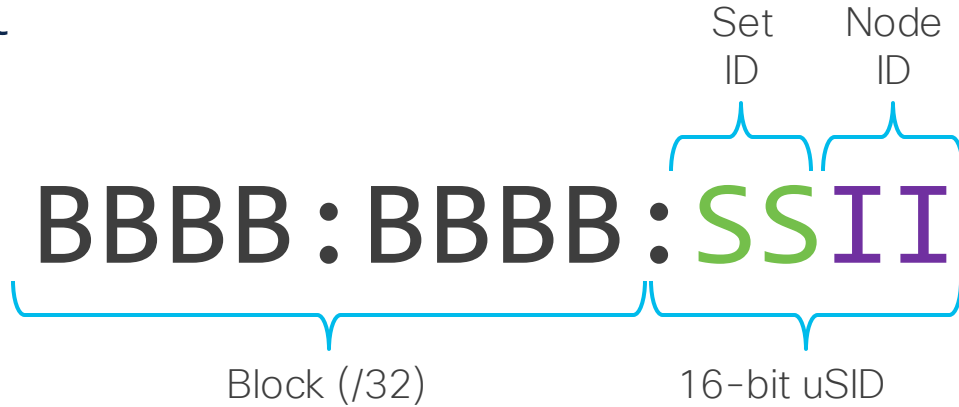
A2: /37₍₂₀₄₈₎ /41₍₁₂₈₎

A3: /39₍₅₁₂₎ /40₍₂₅₆₎

Not Simple

Difficult to plan for growth

Set



- Each block is split to certain number of equally sized Sets
 - /40 is very good example as it follows nibble boundary (/36 possible)
- Each area is assigned certain number of sets
- Summarization is performed on set boundary only
- Single block has 224 /40 sets (because of LIB)

Area addressing – Example with Sets

3 areas A1:350, A2:2100, A3:750 devices

Using Sets

A1: 2x /40

A2: 9x /40

A3: 3x /40

Very Simple and Efficient

Summarization gain 1:256 (if set is /40)

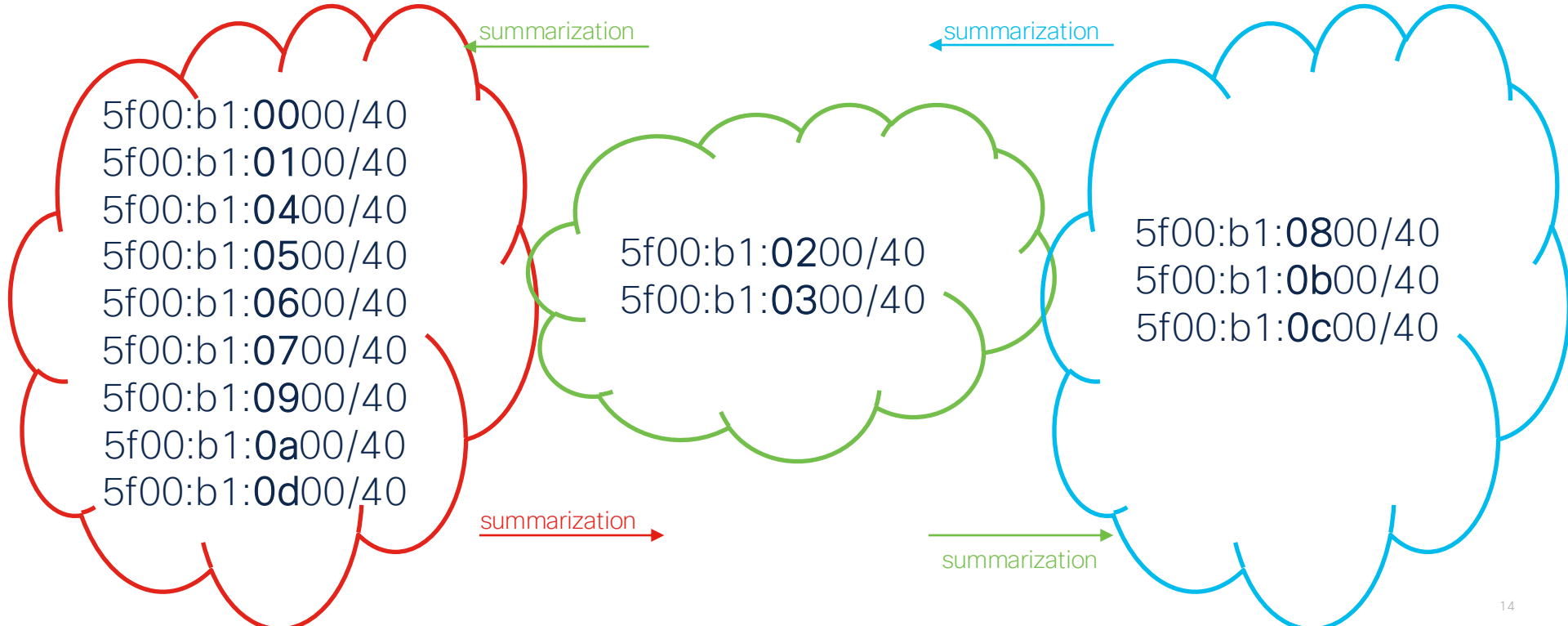
Very simple to grow (sets do not need to be consecutive)

Area addressing – Example with Sets

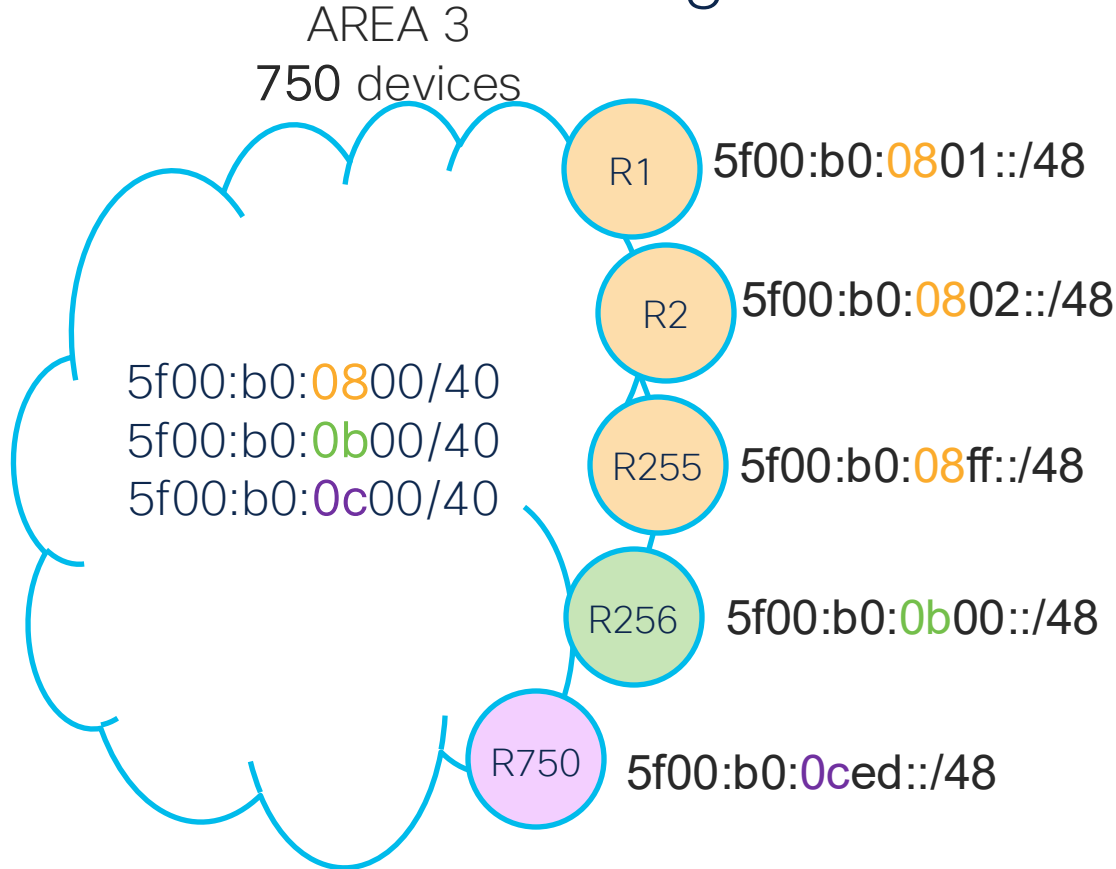
AREA 2
2100 devices

AREA 1
350 devices

AREA 3
750 devices



Locator Addressing in Area 3 – Example with Sets



Flexible Algorithm

- Logical topologies within IGP protocol
 - Include/exclude devices
 - Include/exclude links
 - Can use different metric (IGP, Delay, TE, Generic)
- In SRv6 device needs **independent** locator per algorithm!

Flexible Algorithm locators

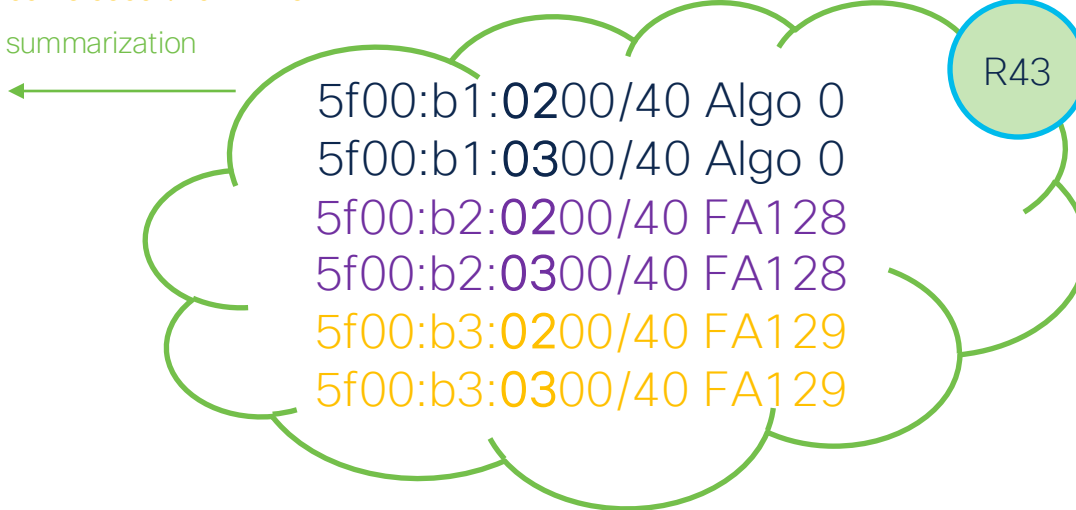
- Always use different block for different algorithm
 - Otherwise LIB is shared-> limited scalability
- Use same Set and Node ID across the algorithms
 - Improves Operational simplicity
 - Same summarization rules for all algorithms
- Number of algorithms in the network will be always limited
 - Current support for 12 ->
 - 1 nibble

BBBB : BBB**A****:****SS****II**

Summarization Example for AREA 1-3 Algorithms

5f00:b1:**02**00::/40 Algo 0
5f00:b1:**03**00::/40 Algo 0
5f00:b2:**02**00::/40 FA128
5f00:b2:**03**00::/40 FA128
5f00:b3:**02**00::/40 FA129
5f00:b3:**03**00::/40 FA129

summarization



Locators for R9 in Set 2

5f00:**b1**:0209::/48 - Default Algorithm

5f00:**b2**:0209::/48 - Flexible Algorithm **128**

5f00:**b3**:0209::/48 - Flexible Algorithm **129**

Large Networks

- Any network which cannot be addressed from single block
 - $\sim > 35k$ devices
- We need multiple blocks
 - Each block is assigned to “region”
- Number of regions should be size of network/35k
 - To minimize number of containers for any TE policy
 - (it is more important than geographical regions)
- Region must be encoded into the block
- Follow nibble boundary
 - Operational simplicity

Large Networks

- 1 nibble for region – 16 regions
- Operation simplicity
- Addressable network size is 16x 35k devices =**500k+** devices!

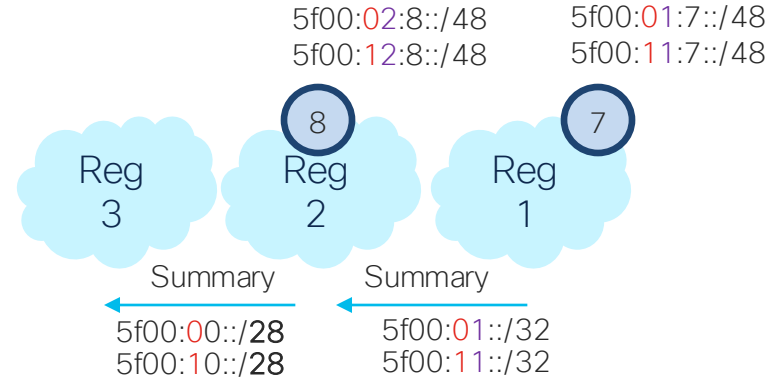
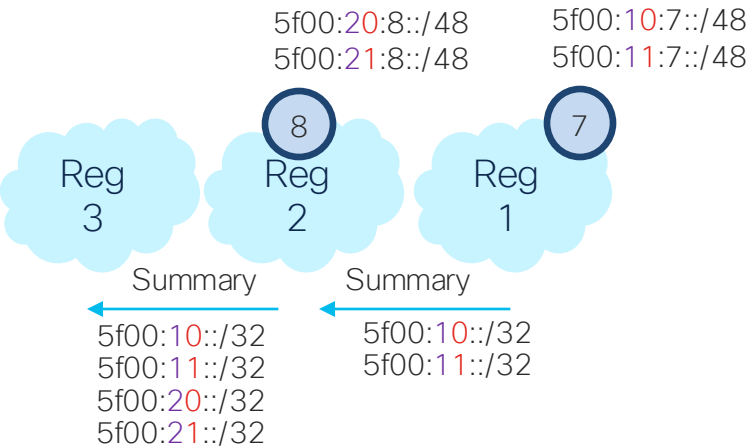
Locator Structure- Large Networks

Now Region- **R** and Flex Algo **A** must be encoded into the block

- BBBB:BB**RA**:SSNN::/48

or

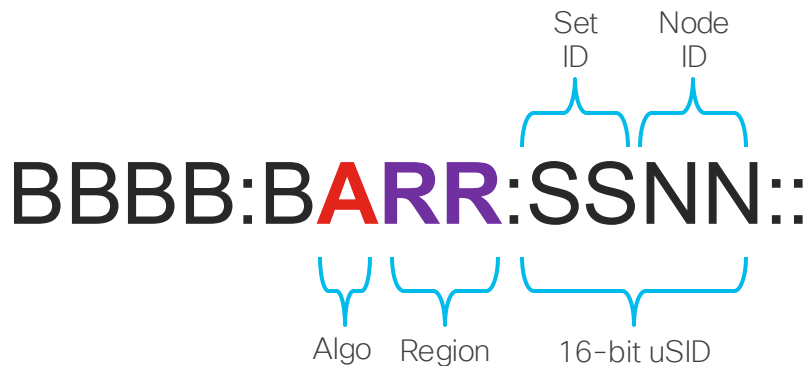
- BBBB:BB**AR**:SSNN::



Different Flex Algo cannot be summarized together!

Very Large Networks

- More than 500k+ device
- We need more than 16 regions (35k each)
- Follow nibble boundary



- 8M+ devices

Loopback Addressing

- Independent



Locator: 5f00:0:3::/48
Lo0: 2001:db8::1/128

- From Locator



Locator: 5f00:0:3::/48
Lo0: 5f00:0:3::1/128

- Smaller LSP
- No loopback addressing planning
- Summarization!!

*Note: only last nibble can be altered!

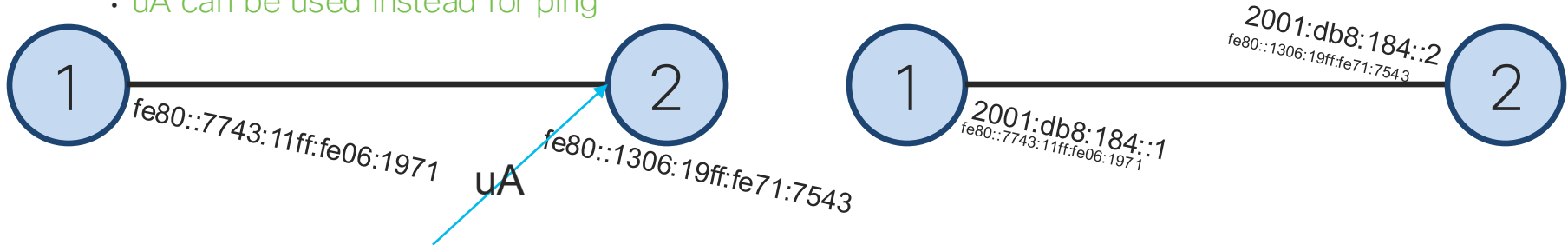
Interface Addressing

- Link Local Only

- no config
- no address planning
- new device insertion
- smaller LSP
- Interfaces not remotely reachable
- uA can be used instead for ping

- Global (ULA/GUA)

- Must be configured
- Must be planned, maintained
- Interfaces remotely reachable



```
RP/0/RP0/CPU0:r1#ping 5f00:b0:2:e000::
```

```
Sun Sep 21 22:42:23.383 UTC
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 5f00:b0:2:e000::, timeout is 2 seconds:
```

```
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/4/7 ms
```

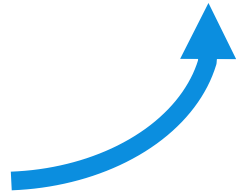
Conclusion for SRv6 Addressing

- Use Private space 5f or ULA (GUA supported)
- Optimize for TE
- <~35k devices – single block per algorithm
- <~500k devices – maximize span of block
- >500k devices – let me know!
- Sets are very helpful to make addressing simple and scalable
- Loopbacks from locators – simplification
- LL only is possibility

Simplicity Always Prevails



Furthermore, with more scale and functionality





The bridge to possible